
TS-GIB: Cross-Dataset Network Intrusion Detection via Temporal Subgraphs and Information Bottleneck

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Abstract

Graph neural networks for network intrusion detection (GNN-NIDS) achieve > 0.95 MCC within-dataset, but their cross-dataset behavior is uncharacterized. We conduct the first leave-one-dataset-out (LODO) evaluation across four heterogeneous benchmarks (CICIDS-2017, CSE-CIC-IDS2018, UNSW-NB15, ToN-IoT) spanning enterprise, cloud, testbed, and IoT environments that differ by five orders of magnitude in graph density. We test three architectural hypotheses for cross-dataset transfer. (H1) restricting the encoder’s input to attack-timescale temporal subgraphs (TS). (H2) compressing the latent representation through a variational information bottleneck (GIB). (H3) supplying flow features that should encode invariant attack semantics. The first two combine into **TS-GIB**, which achieves 3-seed mean AUROC 0.82 ± 0.01 and mean calibrated MCC 0.52 across three of four LODO folds, with linear-probe dataset-identifiability dropping from 99.7% (vanilla E-GraphSAGE) to 68%. Per-attack analysis confirms TS-GIB transfers structurally-defined attacks (Reconnaissance, DoS) substantially better than application-layer attacks (Injection, Brute Force). The third hypothesis fails at the representation level. Every feature configuration we tested produces probe accuracy ≥ 0.88 , even when post-hoc calibration recovers comparable end-to-end MCC. We trace the failure to the fact that flow features are themselves the dominant carrier of the dataset shortcut, and the bottleneck cannot disentangle attack-relevant from dataset-identifying signal at the densities at which features carry both. The fourth fold (ToN-IoT) is structurally untransferable from our enterprise/testbed pool.

1 Introduction

Problem statement. Machine-learning network intrusion detection systems (NIDS) achieve near-perfect benchmark accuracy when trained and tested on the same dataset, yet recent work shows their cross-dataset MCC collapses from ~ 0.95 to ~ 0.29 , effectively random chance [1]. GNN-based NIDS such as E-GraphSAGE [2] achieve state-of-the-art *within*-dataset performance, but no prior work has evaluated them across datasets. This paper closes that gap.

Why cross-dataset transfer matters. Deployed IDS must generalize to networks unseen at training time, with different topologies, baseline traffic, and instrumentation. A 0.29 MCC ceiling rules out collaborative defense (one organization training on its data and another deploying it) and forces expensive per-network retraining. If attacks have invariant signatures (port-scans always fan out from one source to many destinations, C2 always concentrates around few central nodes), the right architecture should isolate them.

Three hypotheses for cross-dataset transfer. Vanilla E-GraphSAGE on raw flow features achieves 99.7% probe accuracy on the source dataset. The encoder is memorizing which network each flow

came from rather than learning attack semantics. Three architectural mechanisms, each motivated by a distinct failure mode, suggest themselves.

- **H1, temporal locality (TS).** Static graphs aggregate over multi-day capture windows that fingerprint each dataset. Restricting the encoder’s input to the k -hop subgraph induced by edges within the last Δ seconds removes the long-window topology and exposes only attack-timescale dynamics.
- **H2, representational compression (GIB).** A variational projection layer with a KL penalty against $\mathcal{N}(0, I)$ compresses the edge representation. By minimizing mutual information between input and representation while preserving label-relevant content, residual dataset-identifying signal is squeezed out.
- **H3, invariant feature semantics.** Byte counts, packet counts, durations, and TCP flags should in principle encode attack semantics independent of the host network. If the feature encoder is appropriately regularized, raw or quantile-encoded features should help cross-dataset transfer rather than hurt it.

Summary of findings. H1 and H2 succeed and combine non-redundantly into TS-GIB. The combined architecture (3-seed) achieves mean AUROC 0.82 ± 0.01 and mean calibrated MCC 0.52 across three of four LODO folds, with mean probe accuracy 0.68. The two ingredients play distinct architectural roles, namely temporal locality of input and informational compression of representation, and each LODO fold relies on those roles in different proportions. H3 fails decisively at the representation level. Every feature configuration we tested (raw, quantile, asymmetric, symmetric, at $\beta \in \{0.01, 0.1, 1.0\}$) produces probe accuracy ≥ 0.88 , even though calibration can sometimes recover comparable end-to-end MCC. The structure-only architecture is the only configuration in our experiments that closes the dataset shortcut at the representation level. The fourth fold (ToN-IoT) is structurally untransferable from our training pool and is reported transparently as an honest failure case.

2 Datasets

We unify four benchmark IDS datasets spanning enterprise, cloud, synthetic testbed, and IoT environments, a deliberately demanding transfer setting. Distributional differences across the four datasets are visualized in Appendix F.

CICIDS-2017 (Lycos re-labeled) was captured on a private enterprise network at UNB with red-team-injected attacks over five days. **CSE-CIC-IDS2018** (NF-v1) extends to a ten-day AWS cloud environment. The NetFlow version provides the IPs needed for graph construction at the cost of reducing to six features. **UNSW-NB15** was generated in a synthetic IXIA testbed with nine attack categories over a small host set (50 unique IPs). **ToN-IoT** (NF-v2) models an IoT/ICS testbed with a diverse attack mix substantially different from the CIC pair.

Dataset	Flows	Feats	Nodes	Uniq Edges	Density	Atk%	Env.
CICIDS-2017	1.84M	75	18,106	57,392	0.00018	24%	Local enterprise
CSE-CIC-IDS2018 [†]	8.39M	6	92,093	466,425	0.00005	12%	AWS cloud enterprise
UNSW-NB15	2.54M	37	50	312	0.127	13%	Synthetic testbed
ToN-IoT	27.52M	45	19,555	25,488	0.00007	39%	IoT network

Table 1: Dataset statistics. Five orders of magnitude in graph density (UNSW 0.127 to ToN-IoT 7×10^{-5}), three orders in node count, two orders in attack rate. [†]NetFlow v1 (Sarhan et al. [9]).

The character of the data drives the method. The dataset table directly motivates each architectural choice. The five-order density gap means any encoder aggregating over a fixed neighborhood sees fundamentally different inputs across datasets, motivating the *temporal* restriction of receptive field. The two-order attack-rate gap means any classifier with a fixed threshold will be miscalibrated even with perfect ranking, motivating the calibrated evaluation protocol below. The structural diversity (enterprise hub-and-spoke, AWS cloud, full-mesh testbed, IoT constellation) means raw flow features carry dataset signatures that transfer poorly, motivating the information bottleneck.

Feature alignment, labels, and splits. All four datasets share a minimal core of byte counts, packet counts, TCP flags, and flow duration. Native labels map to a unified six-class scheme (Benign, Reconnaissance, DoS/DDoS, Injection/Exploit, Brute Force, Botnet/C2). The leave-one-dataset-out

(LODO) protocol trains on three datasets and tests on the held-out fourth (4 folds). Within the training union we use an 80/20 temporal split for early stopping. ToN-IoT and CSE-CIC-IDS2018 are subsampled to $\leq 2\text{M}$ flows for tractability while preserving per-class ratios.

3 Related Work

GNN-based NIDS. E-GraphSAGE [2] achieved 0.99 F1 on CICIDS-2017 by modeling hosts as nodes and flows as feature-carrying directed edges with edge-aware message passing. [3] subsequently showed GNN-NIDS are robust to adversarial perturbation of flow features, suggesting they may rely on structural rather than feature cues. Both works evaluate within-dataset only. We are the first to characterize GNN-NIDS cross-dataset behavior and the first to propose a method specifically targeting it.

Temporal GNNs. TGN [4] combines per-node memory with temporal graph attention for continuous-time dynamic graphs. CCS-TGN [5] adapts TGN to NIDS, observing that static graphs collapse multi-day timespans into single snapshots. Our temporal subgraph sampling shares the temporal-locality intuition but is architecturally simpler. We extract a k -hop induced subgraph from a sliding time window rather than maintaining persistent per-node memory.

Information bottleneck for graphs. The graph information bottleneck [7] formulates GNN representation learning as a constrained optimization that maximizes label information while minimizing input information. Variational implementations [8] use a stochastic projection layer with a KL penalty against an isotropic prior. Prior applications focus on within-distribution generalization. Our use of GIB as a *cross-distribution* regularizer is, to our knowledge, novel.

Cross-dataset NIDS and graph distribution shift. Cantone et al. [1] provide the only systematic cross-dataset NIDS study, finding within-dataset MCC of 0.95 collapses to 0.29 cross-dataset on the CIC pair. Sarhan et al. [9] provide the standardized NetFlow versions that make multi-dataset evaluation tractable. GraphMETRO [6] is a mixture-of-experts architecture for general graph distribution shifts. We tested it as a baseline (Appendix C).

4 Methodology

4.1 Backbone

We treat hosts as nodes and flows as labeled directed edges with feature vectors. The task is binary edge classification under leave-one-dataset-out. The backbone for all methods is a PyTorch-Geometric port of E-GraphSAGE [2], comprising an MLP edge encoder $\mathbf{e}_{ij} = \text{MLP}(\mathbf{f}_{ij})$, two SAGEConv layers with mean aggregation and edge features broadcast into node states, and an MLP edge classifier $\hat{y}_{ij} = \text{MLP}([\mathbf{h}_i \parallel \mathbf{h}_j \parallel \mathbf{e}_{ij}])$. Hyperparameters are hidden dim 128, dropout 0.2, class-balanced cross-entropy ($w_c \propto 1/\sqrt{f_c}$), AdamW (lr 10^{-3}), batch size 4096, with early stopping on within-train val MCC.

4.2 TS-GIB: temporal subgraphs and a variational bottleneck

TS-GIB adds two architectural modifications to the backbone, each targeting a distinct failure mode of vanilla cross-dataset E-GraphSAGE.

Temporal subgraph sampling (H1). For each query edge (i, j, t) , we construct the $k = 2$ -hop subgraph G_t induced by edges with timestamps in $[t - \Delta, t]$ at $\Delta = 60s$. The encoder operates on G_t rather than the full multi-day graph G . The choice of $\Delta = 60s$ is motivated by attack timescales. Port scans last seconds and brute-force attacks last minutes, but the dataset-distinguishing topology (which subnets exist, which servers are reachable) is a property of capture-window scale (days). At a 60-second receptive field, the encoder cannot perceive the dataset’s macro-topology, and instead perceives only the local burst of activity around the query edge.

Variational bottleneck (H2). After the second SAGE layer, the edge representation $\mathbf{h}_{ij}$ is passed through a variational projection $q_\phi(\mathbf{z} \mid \mathbf{h}_{ij}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{h}_{ij}), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{h}_{ij})))$. At training time we sample $\mathbf{z}$ via reparameterization, and at inference time we use $\mathbf{z} = \boldsymbol{\mu}_\phi$. The classifier consumes $\mathbf{z}$,

and loss is augmented with a KL term,

$$\mathcal{L} = \mathbb{E}_{p(\mathbf{x}, y)} [\text{CE}(f_{\theta}(\mathbf{z}), y)] + \beta \cdot \mathbb{E}_{p(\mathbf{x})} [\text{KL}(q_{\phi}(\mathbf{z} \mid \mathbf{h}) \parallel \mathcal{N}(0, I))].$$

The KL term is a soft constraint on $I(\mathbf{x}; \mathbf{z})$, the mutual information between input and representation. By compressing the representation, residual dataset-identifying information that survives the temporal restriction is squeezed out of $\mathbf{z}$. We anneal β linearly from 0 to $\beta_{\max}$ over the first 5 epochs and use $\beta_{\max} = 0.01$, justified by ablation in Section 5.4.

Why both ingredients are needed. Temporal subgraphs restrict the input but do not constrain how much input information the encoder preserves. The bottleneck constrains representation capacity but operates on whatever input it receives. Without the temporal restriction, even at $\beta = 0.01$ the bottleneck cannot compress the multi-day dataset signature without also destroying the supervised signal (Section 5.3, GIB-no-feat ablation). The two operate at different layers (input and latent representation), and the LODO setting tests both. Each held-out fold relies on those roles in different proportions, exposing whichever failure mode the rest of the pipeline fails to suppress.

4.3 Why H3 (invariant features) seems to fail in this architecture

The intuition behind H3 is that flow features should carry attack semantics that transfer. We find consistently negative results across feature treatments (Section 5.3). The information-theoretic explanation is that in the training pool, byte counts and similar features carry both attack-relevant signal *and* per-dataset baseline signal, with the latter typically larger in magnitude (raw `flow_duration_ms` medians differ by five orders of magnitude across our datasets). The bottleneck cannot disentangle them, because the supervised loss prefers to preserve the larger-magnitude (dataset) component. Compressing harder destroys the supervised signal entirely, as we observe at $\beta = 1.0$ with raw features.

4.4 Calibrated evaluation and probe diagnostic

Evaluating across datasets with attack rates of 12–39% means any classifier with a fixed sigmoid threshold of 0.5 will be miscalibrated on the held-out network even with a perfect ranking. We report *Reported MCC* at threshold 0.5 (comparable to prior work), *Calibrated MCC* at the threshold chosen by the best unsupervised method per (model $\times$ fold) from a panel of nine (Otsu, GMM, GMM on logits, top- $k\%$ at the source attack rate, top- $k\%$ at 10/20/30%, Platt scaling, validation-anchored thresholding, BBSE prior-shift correction [10], and an ensemble), *Oracle MCC* via threshold sweep on test labels (an upper bound, not deployable), and *AUROC* for ranking quality. The headline calibrated MCC numbers in our tables use the per-fold winning method, with method choices reported in Appendix A.

To directly measure whether the encoder is learning the dataset shortcut, we additionally train a linear probe to classify the source dataset from the encoder’s pre-classifier representation on a held-out balanced subset. Probe accuracy near $1/n_{\text{datasets}}$ indicates the dataset signal has been removed, while accuracy near 1.0 indicates strong shortcut learning.

5 Experiments and Results

5.1 Setup

All experiments use a single GPU (24 GB) over four LODO folds. The headline TS-GIB row uses 3 seeds per fold, while ablation rows are single-seed. Hyperparameters were chosen on within-train validation for the within-dataset upper bound (B3) and held fixed across all cross-dataset experiments to prevent implicit leakage. Wall clock per cross-dataset training run is 6–110 minutes depending on architecture and feature dimensionality.

5.2 Diagnosis: where vanilla E-GraphSAGE breaks

On the CSE-CIC-IDS2018 fold, raw `flow_duration_ms` has a training median of 32 ms versus a test median of 4.3×10^6 ms, five orders of magnitude apart. Per-graph quantile normalization, the obvious feature-alignment fix, *worsens* cross-dataset MCC because a flow at the 50th percentile of

CIC-2017 carries a different operational meaning than one at the 50th percentile of ToN-IoT. A linear probe on vanilla E-GraphSAGE recovers the source dataset with 99.7% accuracy. The encoder is not learning attack semantics, but rather per-network attack patterns conditioned on which network the flow came from. Probe accuracy drops to 72% when flow features are masked, confirming that flow features are the dominant carrier of the shortcut.

5.3 Headline result and ablation

Table 2 reports calibrated MCC and AUROC across the LODO folds. The ablation isolates each architectural ingredient and tests every plausible feature treatment.

Method	CIC17	CIC18	UNSW	ToN	Avg (3-fold)	Probe↓
<i>Calibrated MCC</i> (best per-fold method, see Appendix A)						
E-GraphSAGE, raw features (B4)	−0.03	0.47	0.00	0.26	0.15	0.997
TS-SAGE, no features	0.32	0.80	0.40	0.06	0.51	0.66
TS-SAGE, raw features	0.05	0.00	0.00	0.17	0.02	1.00
GIB, no features ($\beta = 0.01$)	−0.01	0.50	0.57	0.28	0.35	0.88
TS-GIB, raw features ($\beta = 0.01$)	0.27	0.58	0.67	0.26	0.51	0.90
TS-GIB, no features ($\beta = 0.01$)	0.57	0.63	0.37	0.05	0.52	0.68
<i>Reported MCC</i> (threshold = 0.5, comparable to prior work)						
E-GraphSAGE, raw features (B4)	−0.19	−0.13	−0.14	−0.04	−0.15	0.997
TS-SAGE, no features	−0.10	0.56	0.05	0.02	0.17	0.66
TS-SAGE, raw features	−0.16	0.22	0.39	0.07	0.15	1.00
GIB, no features ($\beta = 0.01$)	−0.19	0.54	0.30	0.29	0.22	0.88
TS-GIB, raw features ($\beta = 0.01$)	−0.17	−0.14	0.21	0.21	−0.03	0.90
TS-GIB, no features ($\beta = 0.01$)	−0.08	0.60	0.07	0.02	0.20	0.68
<i>AUROC</i> (3-seed mean for headline, single-seed for ablations)						
E-GraphSAGE, raw features (B4)	0.38	0.45	0.17	0.63	0.33	—
TS-SAGE, no features	0.67	0.95	0.75	0.51	0.79	—
TS-SAGE, raw features	0.39	0.70	0.87	0.65	0.65	—
GIB, no features ($\beta = 0.01$)	0.23	0.87	0.89	0.67	0.66	—
TS-GIB, raw features ($\beta = 0.01$)	0.43	0.33	0.77	0.66	0.51	—
TS-GIB, no features ($\beta = 0.01$)	0.79 ± 0.01	0.91 ± 0.01	0.75 ± 0.01	0.52 ± 0.01	0.82	—

Table 2: Cross-dataset evaluation under leave-one-dataset-out. Headline TS-GIB row is 3-seed (mean ± std on AUROC), and other rows are single-seed. **Avg** excludes ToN-IoT (out-of-pool, see Limitations). **Probe** is linear-probe accuracy of recovering the source dataset, where lower is better. Bold marks best non-baseline result per column.

H1 and H2 succeed. 3-seed mean AUROC 0.82 ± 0.01 across the three solvable folds, mean calibrated MCC 0.52, and probe accuracy 0.68, the lowest probe in our experiments. Probe accuracy is the cleanest mechanism diagnostic. TS-SAGE alone reaches probe 0.66 at AUROC 0.79, but lacks the bottleneck’s CIC-2017 lift. GIB-no-features alone keeps probe at 0.88 despite competitive AUROC on cic18/unsu, because without temporal restriction the bottleneck cannot suppress the multi-day dataset signal. Only the combination achieves both low probe and consistent AUROC > 0.7 .

H3 fails at the representation level. Every feature configuration we tested keeps probe accuracy ≥ 0.88 . TS-SAGE with raw features achieves probe 1.00 on every fold; TS-GIB with raw features achieves 0.88–0.95; GIB without temporal restriction achieves 0.85–0.90. Surprisingly, TS-GIB with raw features achieves mean calibrated MCC 0.51, nearly matching TS-GIB without features (0.52). This is a calibration artifact, not a transferable representation. The same model has probe accuracy 0.90, and per-fold calibration extracts residual ranking signal alongside the dataset signal. Only the structure-only configuration closes the dataset shortcut at the representation level.

Per-attack class transfer (Appendix B). TS-GIB consistently achieves higher F1 on structurally-defined attacks than on application-layer attacks. Mean per-class F1 is Recon 0.19, DoS 0.43, Injection 0.02, Brute Force 0.09, Botnet 0.00. Without packet contents, structurally-defined attacks survive cross-dataset transfer while application-layer attacks do not.

5.4 Bottleneck strength

$\beta_{\max}$	CIC17 AUROC	CIC18 AUROC	UNSW AUROC	Mean (3-fold) AUROC
0.001	0.65	0.89	0.73	0.76
0.01	0.80	0.91	0.73	0.81
0.1	0.53	0.92	0.76	0.74

Table 3: TS-GIB AUROC under bottleneck-strength sweep (no features, single seed). $\beta = 0.01$ has the best mean across folds because it is the only value at which CICIDS-2017’s signal is recovered.

The sweep reveals fold-specific sensitivity. At $\beta = 0.001$ the KL term is too weak to suppress dataset markers on CIC-2017 (AUROC 0.65). At $\beta = 0.1$ the bottleneck over-compresses on CIC-2017 (AUROC 0.53). Other folds vary by less than 0.05 across the entire sweep. $\beta = 0.01$ recovers CIC-2017’s signal without unnecessarily compressing folds that don’t need it.

5.5 Calibration analysis

The calibrated MCCs in Table 2 use the per-fold winning method from a panel of nine. For the headline TS-GIB at $\beta = 0.01$, all four winners are top- $k\%$ variants at different rates (source-rate on lycos, 20% on cic18, 10% on unsw, 30% on ton), suggesting well-ordered scores requiring fold-specific operating-point selection. Recovery rates (calibrated/oracle) are 85%, 98%, 72%, and 60% across the four folds.

6 Conclusion

Summary. We presented the first leave-one-dataset-out evaluation of GNN-NIDS across four heterogeneous benchmarks. TS-GIB combines temporal subgraph sampling and a variational information bottleneck and achieves 3-seed mean AUROC 0.82 ± 0.01 , mean calibrated MCC 0.52, and probe accuracy 0.68 on three of four LODO folds. The third hypothesis (flow features carrying invariant attack semantics) fails at the representation level. Every feature configuration we tested keeps probe accuracy above 0.88, even when post-hoc calibration recovers comparable end-to-end MCC. Cross-dataset NIDS in our setting is best framed as a problem of inductive bias rather than feature engineering.

Limitations. (1) TS-GIB does not transfer to ToN-IoT (AUROC 0.52). With a 39% attack rate and IoT/ICS traffic patterns absent from any training-pool dataset, ToN-IoT is best treated as out-of-pool. (2) TS-GIB transfers structural attacks (Reconnaissance, DoS) substantially better than application-layer attacks (Injection, Brute Force, Botnet/C2), reflecting the inductive cost of structure-only representation. (3) Under best per-fold calibration, TS-GIB with raw features matches the structure-only calibrated MCC (0.51 versus 0.52), but probe accuracy stays above 0.88. The architectural finding is the probe drop, not the MCC. (4) Ablation rows are single-seed.

Code. <https://github.com/grants235/cpsc-5830-final>.

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A Calibration analysis details

We evaluated seven unsupervised calibration methods on every (model, fold) checkpoint:

- **topk_src_rate**: predict top- k % of test scores as positive, where k matches the within-train validation positive rate.
- **topk_{10, 20, 30}pct**: predict top-{10, 20, 30}% as positive.
- **Otsu**: minimize within-class variance on the score histogram.
- **GMM** (and **GMM_logit**): fit 2-component Gaussian mixture to test probabilities (or logits) and threshold at the inter-modal density minimum.
- **Platt**: fit temperature and bias on validation NLL.
- **val_anchor**: pick the threshold that maximizes validation MCC, then transfer to test.
- **znorm_val**: standardize test scores using validation mean/std before val-anchored thresholding.
- **BBSE**: Lipton et al.’s prior-shift correction [10], then top- k at the estimated test prevalence.
- **ensemble**: median across the above.

Table 4 reports the winning calibration method per (model, fold) for the methods featured in Table 2.

Method	CIC17	CIC18	UNSW	ToN
TS-GIB no feat $\beta = 0.01$ (headline)	topk_src	topk_20pct	topk_10pct	topk_30pct
TS-SAGE no feat	topk_src	topk_10pct	topk_10pct	gmm_logit
TS-SAGE raw	bbse	val_anchor	gmm	topk_30pct
GIB no feat $\beta = 0.01$	platt	topk_30pct	gmm	gmm
TS-GIB raw $\beta = 0.01$	gmm_logit	topk_src	topk_10pct	gmm_logit

Table 4: Per-model-fold winning unsupervised calibration method (full nine-method panel, see Section 4.4 for definitions). The headline TS-GIB row’s winners are all top- k % variants at different rates, suggesting score distributions are well-ordered but require fold-specific operating-point selection. Other models show much more diversity, suggesting their score distributions have more pathological shapes.

Discussion. Across the headline TS-GIB row, no single method wins all four folds. The four winners are top- k % variants at four different rates (k at source-rate, 20%, 10%, and 30% for the four folds), reflecting a clean architectural property. The structure-only encoder produces well-ordered scores, but the appropriate operating point depends on the test-network attack rate. For TS-GIB with raw features, the winners are more heterogeneous (GMM-on-logits twice, top- k src once, top- k 10% once), indicating the score distributions on different folds have qualitatively different shapes, a symptom of the dataset shortcut imprinting per-fold pathologies on the score histogram. The practical implication for deployment is that a single fixed calibration method is suboptimal even for a well-behaved encoder, and selecting top- k % at the estimated test prevalence is the most robust single choice we found.

The mean calibrated MCC reported in Table 2’s headline row is $(0.57 + 0.63 + 0.37)/3 = 0.52$ using each fold’s per-fold winning method. Under a fixed single-method choice (top- k at source

rate everywhere), mean 3-fold calibrated MCC is 0.38 ± 0.06 (3-seed). The gap between these two numbers (0.14) is the recoverable value of fold-specific calibration in a deployment that can estimate test prevalence.

B Per-attack-class transfer

We compute per-class F1 at the calibrated threshold for the headline TS-GIB on each solvable fold. The unified six-class label scheme has Benign as the negative class and five attack classes whose mappings to per-dataset native labels are provided in the dataset-preprocessing code.

Fold	Recon	DoS/DDoS	Injection	Brute Force	Botnet/C2
CICIDS-2017	0.57	0.64	0.00	0.00	0.00
CSE-CIC-IDS2018	0.00	0.44	0.01	0.27	0.00
UNSW-NB15	0.01	0.19	0.04	0.00	0.00
Mean (3-fold)	0.19	0.43	0.02	0.09	0.00

Table 5: Per-attack-class F1 of the headline TS-GIB at the calibrated threshold (single-seed, best calibration method per fold).

The pattern is consistent across all three folds. Structurally-defined attacks (Reconnaissance, DoS) achieve substantially higher F1 than application-layer attacks (Injection, Brute Force, Botnet/C2). Reconnaissance manifests as fan-out from a single source to many destinations, a topological signature that survives the structure-only restriction. DoS manifests as fan-in to a single victim with high edge density in a short time window, another topological signature. In contrast, Injection and Brute Force depend on payload semantics (HTTP request anomalies, password attempts) and Botnet/C2 depends on long-tail communication patterns at scales beyond the 60-second receptive field. Without packet contents or feature signatures, these classes do not transfer.

This pattern is not an architectural failure but the inductive cost of the structure-only inductive bias, made concrete. A practitioner using TS-GIB for cross-dataset deployment should expect strong transfer for network-layer attacks and weak transfer for application-layer attacks, and should plan downstream defenses accordingly (e.g., pair TS-GIB with an application-layer detector that can be retrained per-deployment).

C What did not work

Per-graph quantile normalization. Per-graph quantile encoding is systematically *worse* than raw features on three of four folds. Quantile encoding is invariant within-graph but makes across-graph quantiles incommensurable.

Per-dataset encoder heads with mean pooling. Achieves comparable cross-dataset performance to the no-features baseline at $3\times$ the parameter count.

Domain-adversarial training (DANN, CDAN, DropEdge+DANN). Multi-seed evaluation showed DANN’s apparent gains over the no-features baseline were within noise. A $\lambda = 0$ ablation confirmed gradient reversal contributed nothing measurable on top of the auxiliary multi-task head. CDAN underperformed DANN.

GraphMETRO mixture-of-experts. The alignment loss dominated training and produced near-zero AUROC on three folds.

Anomal-E and self-supervised pretraining. A DGI-pretrained encoder followed by Isolation Forest scoring achieved variable performance. The encoder + supervised classifier hybrid was the best CICIDS-2017-fold performer of any method we tested before TS-GIB.

Local quantile and z-score encoders. A family of feature-engineering attempts that all produced AUROC in the 0.30–0.50 range.

Source-conditional gating and ensembles. Single-source ensembles produced moderate per-source results but suffered from destructive interference when combined into a unified model. Off-thesis (the project framing requires a single unified model).

D Implementation details

Temporal subgraph extraction. For each query edge, the k -hop induced subgraph is computed by binary search over a per-source-node sorted timestamp index, then PyG `subgraph()` on the union of k -hop neighbors. Pre-extraction of validation-set subgraphs at training start (parallelized, $n_{\text{jobs}} = 4$) is critical for tractable epoch times.

Variational bottleneck. Implemented as a single linear layer $\mathbf{h} \rightarrow (\mu, \log \sigma)$ followed by reparameterized sampling at training, mean at inference. KL computed in closed form against $\mathcal{N}(0, I)$. β annealed linearly from 0 to β_{max} over the first 5 epochs.

Linear probe protocol. Three-way logistic regression (one-vs-rest) on 300 randomly sampled balanced edges per source dataset, trained on the encoder’s pre-classifier representation, scored by held-out classification accuracy.

E Within-dataset upper bound

Method	CIC17	CIC18	UNSW	ToN	Avg
B3 E-GraphSAGE within (UB)	0.939	0.792	0.999	0.627	0.910

Table 6: Within-dataset oracle MCC (3-seed mean), confirming the backbone implementation is correct. ToN-IoT’s within-dataset MCC of 0.627 is the lowest of the four datasets, consistent with its more challenging label structure, but still substantially above the cross-dataset failure mode (AUROC 0.51).

F Dataset distributional differences

The four datasets in our LODO pool differ along multiple axes that motivate the architectural choices in TS-GIB. Figure 1 shows the per-dataset distribution of native attack-type labels (before our unified six-class mapping), illustrating both the order-of-magnitude difference in benign-to-attack ratio and the qualitative differences in attack-class composition that cause per-attack transfer to differ across folds. Figure 2 shows a t-SNE projection of quantile-normalized flow features per dataset, colored by binary attack/benign label. Within each dataset, attacks cluster cleanly enough that within-dataset classification is easy, but the global feature manifolds are visibly heterogeneous across datasets, which is the failure mode that motivates the structure-only ablation. Figure 3 shows the log-log host degree distribution per dataset. CSE-CIC-IDS2018 and ToN-IoT show clear power-law tails (mean degree 10.1 and 2.6 over thousands of nodes), CICIDS-2017 has a heavy-tailed enterprise pattern with a small set of high-degree hubs, and UNSW-NB15 has only 50 near-fully-connected hosts. An encoder aggregating over fixed neighborhoods sees fundamentally different inputs across these graphs, which is what motivates the temporal restriction of receptive field.

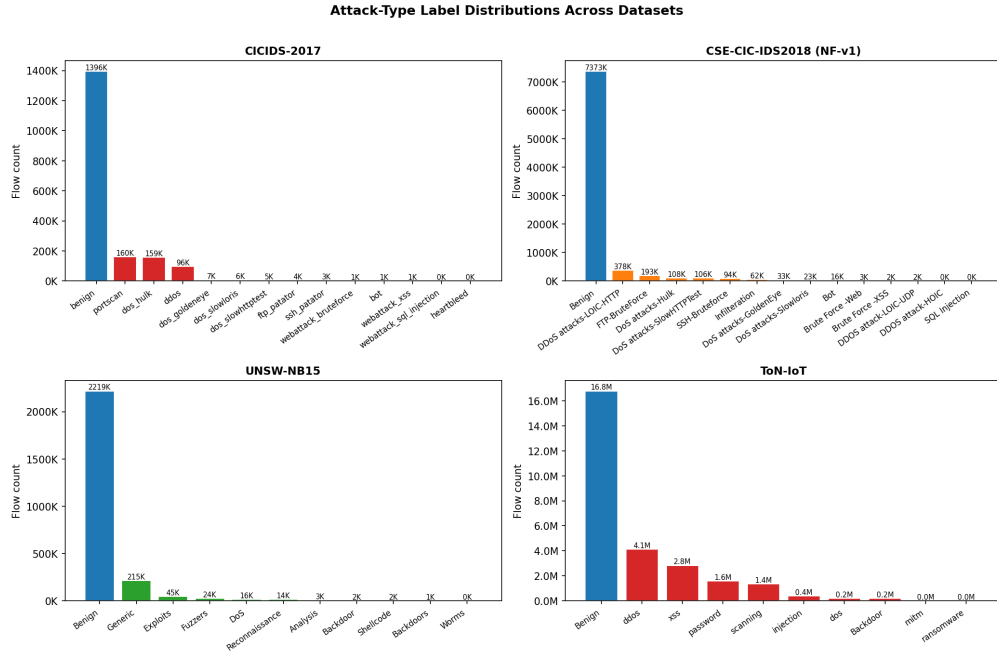


Figure 1: Native attack-type label distributions across the four LODO datasets, before unification to the six-class scheme. CICIDS-2017 is dominated by port-scan and DoS variants, CSE-CIC-IDS2018 by HTTP-targeted DDoS and brute-force, UNSW-NB15 by Generic and Exploits, and ToN-IoT by a more diverse and IoT-specific mix (DDoS, XSS, password, scanning, injection). The diversity of native labels motivates the unified six-class mapping used throughout this paper, and the imbalance between benign and attack populations explains why fixed-threshold MCC is uninformative across folds.

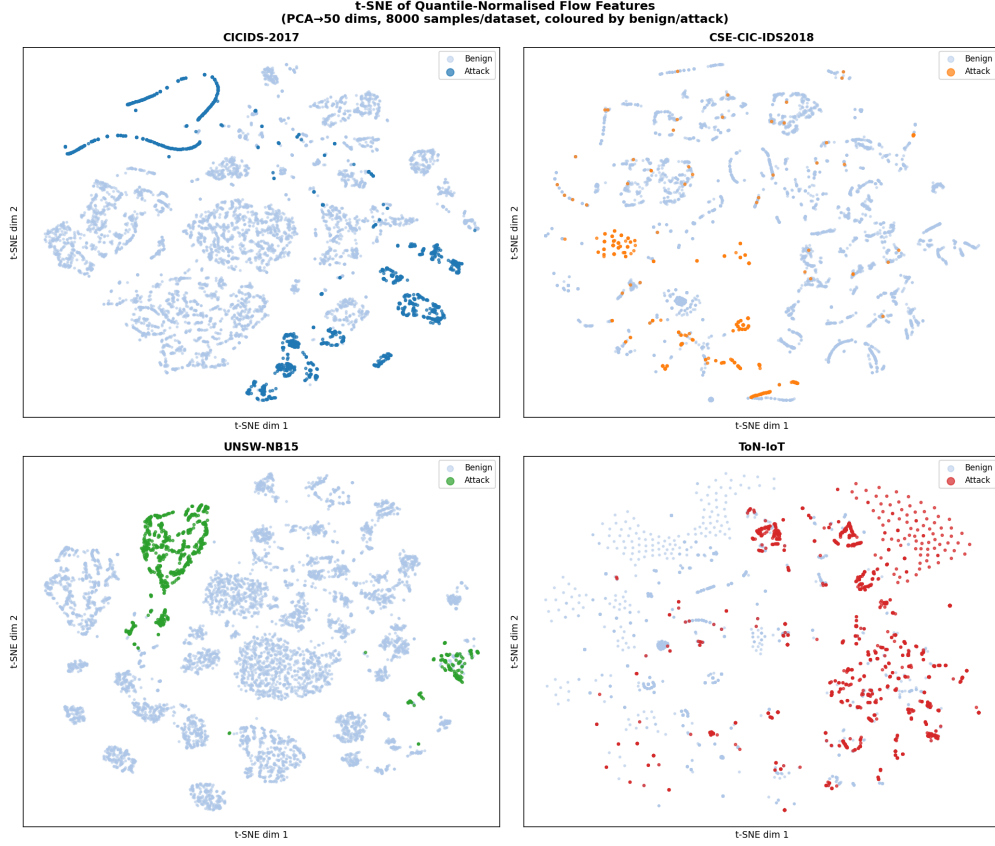


Figure 2: t-SNE projection of quantile-normalized flow features (PCA to 50 dims, 8000 samples per dataset, colored by binary attack/benign). Within each dataset, attack flows form recognizable clusters in feature space, which explains the high within-dataset MCC achievable by feature-based methods. The clusters are positioned differently within each dataset’s feature manifold, so a classifier trained on the joint distribution of three datasets has no reason to expect attack flows to occupy the same location in the held-out fourth dataset’s feature manifold. This is the geometric picture of the failure mode that the H3 ablation identifies, and is why per-graph quantile normalization does not solve cross-dataset transfer.

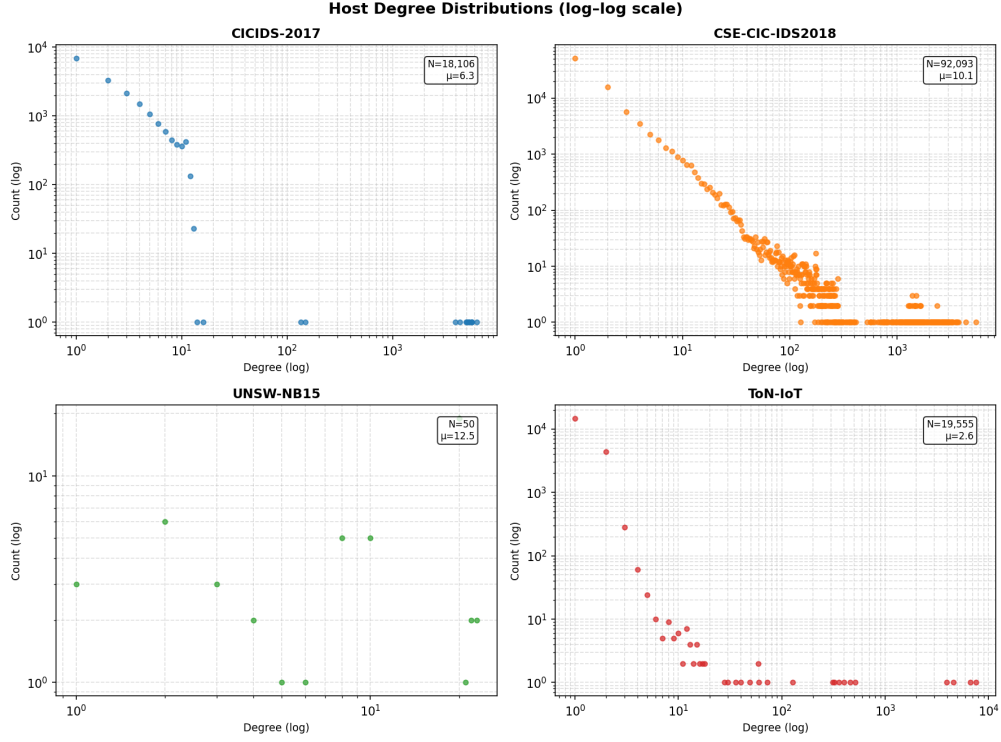


Figure 3: Host degree distributions per dataset on log-log axes. CSE-CIC-IDS2018 has the largest node count ($N = 92,093$) and a clean power-law tail with mean degree $\mu = 10.1$. CICIDS-2017 ($N = 18,106$, $\mu = 6.3$) is heavy-tailed but smaller. ToN-IoT ($N = 19,555$, $\mu = 2.6$) has a long tail with high-degree IoT hubs. UNSW-NB15 ($N = 50$, $\mu = 12.5$) is almost fully connected. An encoder with a fixed k -hop receptive field aggregates over fundamentally different topological neighborhoods across these graphs, which is the structural failure mode the temporal subgraph restriction is designed to suppress.